



Parktown Boys' High School

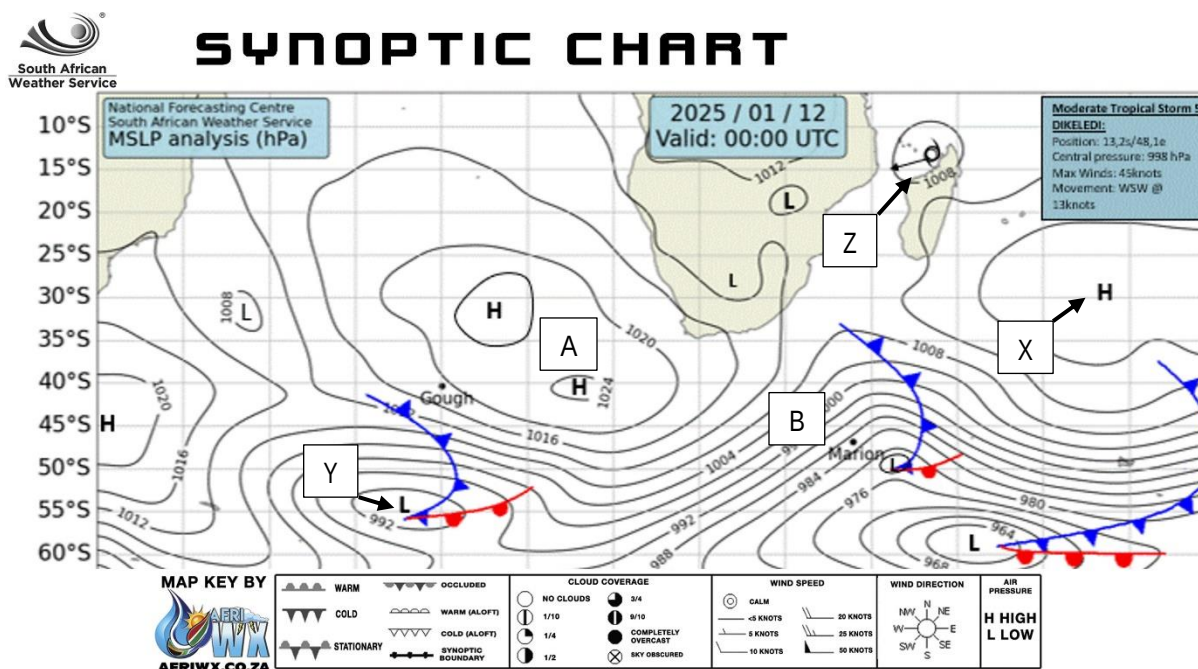
Geography Department

Grade 12 Control Test

MEMO

Question 1 – Short Questions [15]

- 1.1. Refer to the diagram below showing a valley. Answer the questions that follow by choosing only the correct term in brackets. E.g. 1.9. Updog

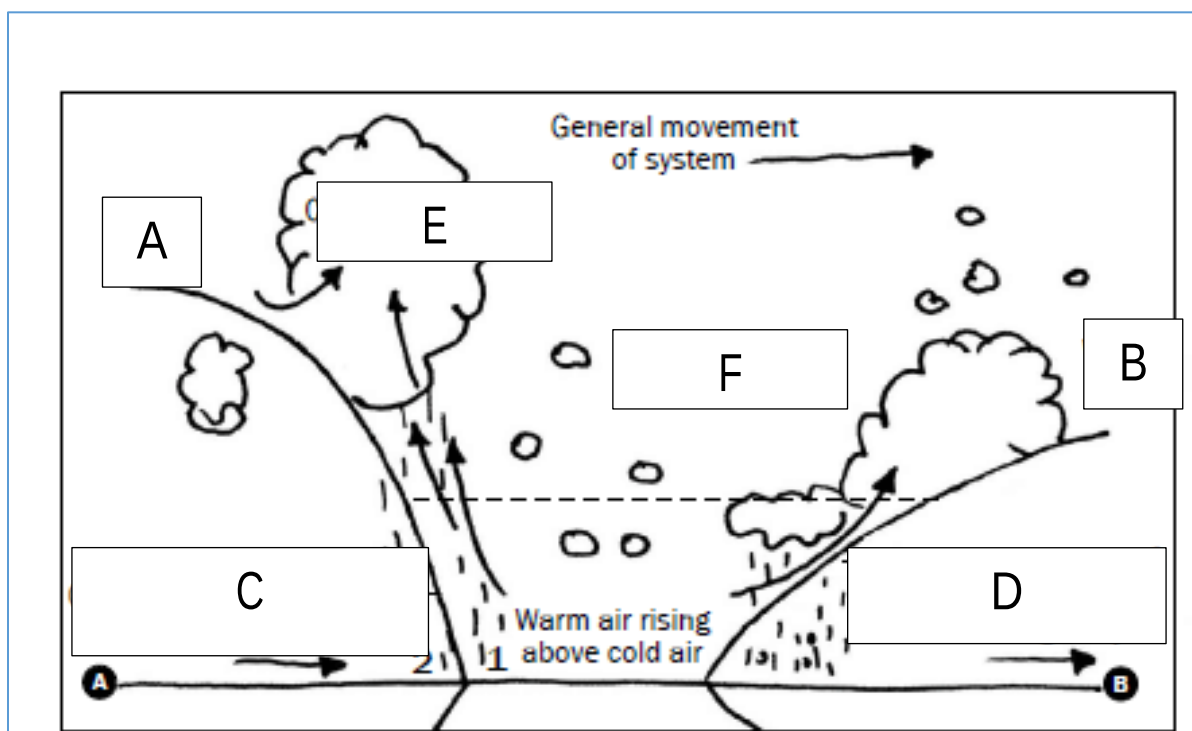


- 1.1.1. The season shown in the synoptic chart is (Summer/Winter/Autumn/Spring).
- 1.1.2. The high pressure at X is the (South Atlantic/South Indian/Kalahari) high pressure.

- 1.1.3. (Z/Y) is a mid-latitude cyclone.
- 1.1.4. (Z/Y) is a tropical cyclone.
- 1.1.5. The wind speed at B is fast. We know this because it has (a steep pressure gradient/Coriolis force is powerful here/it has both a strong pressure gradient force and a strong Coriolis impact).
- 1.1.6. The conditions at A are likely (clear and calm/wet and windy/cold and humid).
- 1.1.7. Tropical storm Dikeledi is the (2nd/3rd/4th) cyclone of the season.
- 1.1.8. Tropical storm Dikeledi, according to the chart, is currently over (Mozambique/Madagascar/South Africa).

(8x1) (8)

- 1.2. Refer to the figure below and statements that follow. Select only the correct word in brackets in each statement to make the statement TRUE.



- 1.2.1. The front at B is the (warm / cold / occluded) front.
- 1.2.2. The temperatures of the air found at C, F and D are (cool, cold and warm / cold, warm and cool / warm, cold and cool) respectively.
- 1.2.3. The symbol for the front found at A is (semi-circles / triangles / a combination of semi-circles and triangles).

- 1.2.4. The clouds that typically form at the front found at A are (cumulonimbus / stratus / cirrus) clouds.
- 1.2.5. The clouds that typically form at the front found at B are (cumulonimbus / nimbostratus / cirrus) clouds.
- 1.2.6. The pressure found at E is (low / high / no) pressure.
- 1.2.7. Once the front labelled A passes over a point, the pressure in that area will (increase / decrease / remain the same).

(7x1) (7)

Question 1.1

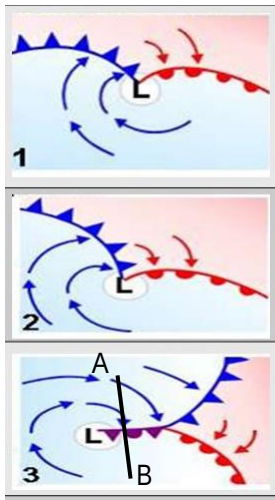
1	Summer
2	South Indian
3	Y
4	Z
5	Strong pressure gradient (Coriolis only affects direction and is NA here)
6	Clear and calm
7	4th
8	Madagascar

Question 1.2

1	Warm
2	Cold, warm, cool
3	Triangles
4	Cumulonimbus
5	Nimbostratus
6	Low
7	Increase

Question 2 – Mid-latitude Cyclones [15]

Refer to the diagram below and answer the questions that follow.



2.1. Provide the labels for the stages found at 1, 2 and 3 respectively. (3x1) (3)

Initial/Developmental ✓ Mature ✓ Occluded ✓

2.2. Draw a fully labelled cross section from A to B in stage 3. This is a cold front occlusion. Make sure you have the following labels:

Heading

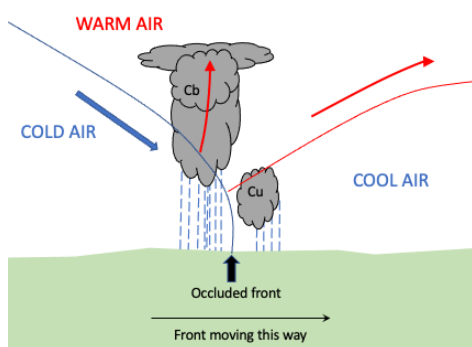
Front symbols

Temperature of air pockets

Cloud types

(4x1) (4)

[Accept reasonable heading] Cross section of a Cold Front Occlusion ✓



Front symbols (Needs triangles on CF, semis on WF and combo on OF) ✓

Cold, warm, cool/cold pockets of air ✓

Cumulonimbus clouds ✓

- 2.3. Why are mid-latitude cyclones more common in winter along the southern coast of South Africa? (1x2) (2)

[Accept any ONE]

Northern hemisphere is receiving more heat and so ITCZ migrates north ✓✓

South Atlantic and South Indian HPs migrate north ✓✓

Polar front migrates north ✓✓

Westerly wind belt shifts north ✓✓

- 2.4. Discuss TWO effects that a mid-latitude cyclone would have on farmers in the Western Cape. (2x2) (4)

[Accept any TWO – MUST BE DISCUSSION]

Increased Rainfall: Mid-latitude cyclones bring heavy rainfall, which can benefit farmers by replenishing water supplies and improving soil moisture for crops. ✓✓

Cold Temperatures and Frost: The passage of the cold front can lead to a drop in temperature, which may damage sensitive crops and affect livestock health. ✓✓

Strong Winds: High winds associated with mid-latitude cyclones can damage crops, uproot trees, and destroy farming infrastructure such as greenhouses and irrigation systems. ✓✓

Flooding: Excessive rainfall can lead to flooding, which may cause soil erosion, waterlogging of crops, and loss of livestock. ✓✓

Delayed Harvesting: Persistent rain and wet conditions may make it difficult for farmers to access fields, delaying harvesting and affecting crop quality. ✓✓

Improved Water Storage: Rainfall from mid-latitude cyclones can help fill dams and reservoirs, ensuring sufficient water supply for irrigation during drier months. ✓✓

- 2.5. Explain what causes a mid-latitude cyclone to move from the south west towards the east of South Africa. (1x2) (2)

Westerly Winds: Mid-latitude cyclones are driven by the prevailing **westerly wind belt**, which moves from **west to east** in the mid-latitudes. ✓✓

Pressure Gradient Force: The difference between high-pressure systems (subtropical high) and low-pressure systems influences the **eastward movement** of the cyclone. ✓✓

Question 3 – Tropical Cyclones [15]

Refer to the write-up of Tropical Cyclone Dikeledi below and answer the questions that follow.

Disaster description

On 11 January, tropical cyclone DIKELEDI passed over the far northern Madagascar, on the following day it passed south of Mayotte (France) and Comoros. On 13 January at 6.00 UTC its centre was located offshore over the Mozambique Channel, approximately 75 km east of the coast of the Nampula province, northeastern Mozambique, with maximum sustained winds of 137 km/h. Media report, as of 13 January, three fatalities, and around 15,200 displaced people across northern Madagascar. In addition, the Interdepartmental Crisis Management Operational Centre (COGIC) reports approximately 20,000 evacuated people across Mayotte. DIKELEDI is forecast to pass along the coast of the Nampula province on 13 January, with maximum sustained winds up to 155 km/h. After that, it is expected to continue moving southward over the Mozambique Channel on 14-15 January, further strengthening. Over the next 96 hours, very heavy rainfall, strong winds and storm surges are forecast over most of Madagascar. The same conditions are also forecast over the next 48 hours for Mayotte, Comoros, and the central-northern coast of Mozambique. ([ECHO, 13 Jan 2025](#))

After its passage on 11 January over northern Madagascar, national authorities report that the death toll stands at three people, 352 people have been displaced and 5,803 affected across Analanjirofo, Diana and Sava regions, northern Madagascar.

Initial reports indicate that more than 7,000 people have been affected by Tropical Cyclone Dikeledi in Madagascar. Over 1,000 houses have been damaged, 203 destroyed. Schools resumed on 13 January in some areas in the north of Madagascar. However, school courses for over 8,000 children have been disrupted as nearly 250 classrooms were damaged or destroyed by winds.

- 3.1. Which ocean did Cyclone Dikeledi pass through while moving towards Mozambique? (1x1) (1)

Indian Ocean ✓

- 3.2. What was the maximum sustained wind speed recorded as Cyclone Dikeledi approached the Nampula province in Mozambique? (1x1) (1)

137 km/h (must have units! 137 is incorrect). ✓

- 3.3. What will happen to Cyclone Dikeledi after it moves into the southern Indian Ocean? (1x1) (1)

It will die out/dissipate

- 3.4. Define the term storm surge and explain why it is a significant hazard associated with Cyclone Dikeledi. (2x2) (4)

Definition of Storm Surge: A storm surge is an abnormal rise in sea level caused by strong winds and low pressure during a tropical cyclone. √√

Why It's a Significant Hazard with Cyclone Dikeledi:

[Any ONE – must be an explanation]

Coastal Flooding: The surge can cause severe flooding in low-lying coastal areas, damaging homes and infrastructure. √√

Erosion and Property Damage: Powerful waves and rising water levels can lead to coastal erosion and destruction of buildings. √√

Loss of Life and Displacement: Floodwaters can be life-threatening, forcing evacuations and displacing communities. √√

Disruption of Transport and Economy: Roads, ports, and businesses can be severely impacted, affecting local economies and emergency response efforts. √√

- 3.5. Discuss TWO economic impacts that Dikeledi would have on Madagascar, and provide TWO solutions that Madagascar could implement to reduce the impact of future cyclones. (4x2) (8)

[Any TWO – Must be ECONOMIC impacts! Can accept other reasonable answers]

Economic Impacts:

Destruction of Infrastructure: Roads, bridges, and communication networks may be damaged, disrupting trade and daily economic activities. √√

Agricultural Losses: Heavy rains and strong winds can destroy crops and kill livestock, leading to food shortages and loss of income for farmers. √√

Business Disruptions: Flooding and storm damage can force businesses to close, resulting in job losses and reduced economic productivity. √√

Tourism Decline: Damage to coastal resorts and transport infrastructure can discourage tourists, leading to revenue losses in the tourism sector. √√

[Any TWO – Can accept other reasonable answers]

Solutions:

Improved Disaster Preparedness: Strengthening early warning systems and evacuation plans can help protect lives and reduce economic losses. ✓✓

Stronger Infrastructure: Constructing cyclone-resistant buildings, roads, and flood barriers can minimize damage during future storms. ✓✓

Reforestation and Coastal Protection: Planting mangroves and preserving natural barriers can help reduce the impact of storm surges and flooding. ✓✓

Financial Support for Affected Communities: Establishing emergency funds and insurance schemes can help businesses and farmers recover more quickly. ✓✓

Question 4 – Application [15]

Read the cartoon below and answer the questions that follow.



4.1. What is 'climate denial' as illustrated in the cartoon? (1x2) (2)

[Concept]

Refusal to acknowledge or believe in climate change/ ✓✓

4.2. What is the name given to these storms when they develop over the Pacific Ocean? (1x1) (1)

Typhoon ✓

4.3. How many hurricanes, including Milton, have occurred in this hurricane season? (1x2) (2)

13 ✓✓

- 4.4. How does the cartoonist use irony to emphasize the impact of climate change? (2x2) (4)

[Accept reasonable answers]

The building is a newspaper/news station ✓✓ It is being blown away by a massive hurricane yet refusing to talk about climate change ✓✓

It is ironic because news stations are supposed to report the truth ✓✓ , and despite being blown away they are still denying climate change ✓✓

It is ironic because the thing they are denying (climate change) ✓✓ , is exactly what is destroying them ✓✓

- 4.5. What role do media and politicians play in shaping public opinion about climate change, as suggested by the cartoon? (2x2) (4)

[Accept reasonable answers – Any TWO]

Spreading Awareness: The media informs the public about climate change by reporting on scientific research, extreme weather events, and global policies. ✓✓

Misinformation or Bias: Some media outlets may downplay or exaggerate climate change issues, influencing how seriously people take the problem. ✓✓

Policy and Legislation: Politicians influence climate action through policies, funding for renewable energy, and international agreements. ✓✓

Political Spin and Denial: Some politicians mislead the public for political gain, either supporting or denying climate change based on their interests. ✓✓

- 4.6. Hurricane Milton became the 5th largest hurricane ever recorded on planet Earth. Explain how rising sea surface temperatures contribute to the formation of stronger hurricanes like Hurricane Milton. (1x2) (2)

[Any ONE – Must be an explanation]

Increased Evaporation: Warmer sea surface temperatures cause more evaporation, leading to higher humidity and greater energy for storm formation. ✓✓

Stronger Winds and Intensification: The extra heat energy fuels the hurricane, causing lower pressure, faster wind speeds, and heavier rainfall. ✓✓

GRAND TOTAL: 60 MARKS